Research Statement

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I am an economist who studies how producers react to regulation. In particular I am interested in how regulation can shift profits from less productive to more productive firms. I have used tools from macroeconomics, theory, applied micro and industrial organization, and applied them to markets like food, education and housing.

1. Regulatory Capture

Large firms lobby for regulation that raises production costs, including their own. In my job market paper 'Regulatory Capture via Relative Prices', I propose that they are seeking a change in relative prices. When a regulation affects some firms proportionately more than others, every price rises, but the least burdened firms raise theirs the least. They become relatively cheaper, attract demand, and gain revenue, profit and market share. Despite being intuitive, this mechanism has not been previously studied in the literature. When a regulation that increases marginal costs unevenly is studied, it is done through the homogeneous-good partial-equilibrium lens of *Raising Rivals' Costs*. But homogeneous goods have a single price. When a regulation that affects firms disparately is studied in differentiated goods, it is done through the fixed-cost selection mechanism of Melitz. But firm exit does not change the relative prices of the surviving firms.

I start by showing that in partial equilibrium, for a broad class of homothetic demand systems, the marginal cost relative to the price aggregator is the sufficient statistic that determines firm outcomes: profit, revenue and market share. This implies that a regulation that increases everyone's costs can still benefit some firms as long as their marginal cost increases proportionally less. I find evidence of this mechanism in the Food Safety Modernization Act (FSMA), the largest food safety reform since 1938. The act brought dry, ready-to-eat foods under preventive controls for the first time. The largest food manufacturers already ran similar controls because exporting required them, and they lobbied for the act. For smaller manufacturers, the act implied new personnel and new programs. Using Nielsen scanner data, I find that large firms grew with respect to smaller firms in products where FSMA binds, but not in similar unaffected placebo categories. For example, in oil-roasted nuts the estimated effect implies that two firms with a revenue ratio of 10:1 ended up with a revenue ratio of 30:1 post regulation.

I then bring the mechanism to general equilibrium in a Nested CES setting. Under the standard assumption that demand across industries is inelastic, a cost-increasing regulation brings expenditure *into* the regulated industry. So, the mechanism *strengthens*: the firms that gained in partial equilibrium gain even more. I use this general equilibrium model and the estimated effects to study the welfare cost of the regulation, and find that the regulation costs at least $362 million across the 11 categories where FSMA binds. Finally, I close by showing that in a simple *Raising Rivals' Costs* homogeneous-goods model, a researcher studying this regulation would conclude instead that welfare increased in 4 of the categories.

This paper shows that we should change how cost-raising regulation is evaluated. A regulation can be universal in law and still pick winners and losers. The burden schedule is itself an object of political economy that changes the size distribution of firms, and its consequences should be measured alongside the regulation's potential benefits.

2. Education

The school voucher literature mostly focuses on student outcomes such as standardized test scores. In some settings the literature finds that outcomes improve and in others that the effect is null or even negative. In 'Supply-side Responses to Targeted Vouchers' (with Cristián Sánchez), we posit that to explain this heterogeneity we need a better understanding of how the producers of education, the schools, react to vouchers.

We study the 2008 Chilean Targeted Voucher reform, where vouchers were offered only to low-income students. The reform was nationwide, but it affected each municipality with varying degrees of intensity. We define the intensity of treatment by the share of low-income students in the municipality, under the logic that a municipality with no low-income students would not be impacted, while a municipality with only low-income students would instead receive the full impact.

Using an event-study design, we estimate the effect on both extensive (entry and exit) and intensive (production inputs and pricing) margins. We find that private schools enter and exit the market more frequently, and make costly efforts to improve education by investing in infrastructure and by improving the teachers' working conditions. Public schools move generally in the same direction, but they do so at a slower pace. Interestingly, we find that the average tuition a low-income student faces *increases*, narrowing the options of the families that the reform was meant to help. On the other hand, the average tuition a high-income student faces *decreases*.

In 'Price and Market Segmentation Consequences of Targeted Vouchers' (with Emiliano Ramírez and Cristián Sánchez), we further explore this response on tuitions. Participation in the voucher program was optional for private schools. The only condition imposed on participating was that they were not allowed to charge a positive tuition to low-income students. This effectively introduced three different types of prices a student would face: in participating schools a zero tuition for low-income students and a positive one for high-income ones, and in non-participating schools a single positive tuition for everyone. The effects on tuitions imply market segmentation and can be explained through the lens of peer-effects where low-income students generate disutility on the families' preferences. As the share of low-income students increases in a participating school, they have to lower tuition to attract high-income students. And as the share of low-income students in the market increases, non-participating schools can charge higher tuition for the privilege of a curated high-income peer group.

To support this argument, we estimate an industrial organization education demand model, and find that high-income students are willing to pay ∼$210 per month to avoid a peer group that consists of only low-income students. This magnitude is ten times the average tuition that high-income students face.

3. Housing

Housing costs as a share of income have been increasing for decades in high-productivity places like Coastal California. A popular explanation is that housing production is inelastic, and that restrictive regulation is in part the reason why. To address the housing costs, the California government approved a suite of state-wide reforms in 2017 that eliminated the burdensome red tape around the permitting of Accessory Dwelling Units (ADUs, small one-unit additions usually built in a backyard). In 'Who Wins With Up-Zoning?' (with Phil Hoxie), we exploit a unique regulatory discontinuity to study the impact of the reforms. There is a ∼1km-wide strip on the

coast of California where the ADU reforms are superseded by the 1976 California Coastal Act. This act imposes prohibitive bureaucratic costs in the permitting of new housing units. Thus, we use a difference in discontinuities design to estimate the causal effects of the ADU reforms on permitting activity, land value, transactions and property taxes.

Our main finding is that the newly acquired ability to build an ADU increased land values by 7%, whether the owners decide to build or not and despite the modest 30% increase in permitting. The option to build is capitalized into land values because it expands the plot's housing capabilities, even if the option is not exercised. To rationalize the increase in land values despite low permitting activity, we build an open-city model with property taxes and endogenous building decisions. The model suggests that property tax shelter, the extent to which longstanding incumbent owners are taxed on a basis substantially below market value, dampens transactions and in turn mutes ADU construction.

The housing industry is directly affected by a large and diverse set of regulations, from zoning and land use to taxes and building codes. Our results suggest that making permitting easier is not enough to incentivize housing construction, and we conclude that we also need to consider the other regulations surrounding this industry.

My research agenda speaks to the importance of regulation in how it shapes and changes producers. Going forward, I plan to keep studying this relationship.